

12 A
$$g = \frac{GM}{R^2} = \frac{G}{R^2} \left(\frac{4}{3} \pi R^3 \rho \right) = \frac{4\pi G}{3} \rho R$$

Hence,
$$\frac{g_M}{g_E} = \frac{\rho_M R_M}{\rho_E R_E} = 0.50 \times 0.70 = 0.35$$

Therefore,
$$\frac{W_M}{W} = \frac{mg_M}{mg_E} = 0.35$$

13 A When the platform is above the equilibrium level and moving down